

Introduction & Hook

Ask students:

“How does a VR system detect when your hand touches or collides with a virtual object without checking every triangle of the object?”

Before checking detailed geometry, VR systems use **simplified shapes called bounding volumes**. And collision optimization using bounding box.

Today we combine:

- Transformations
- Object positioning
- Collision optimization

PART 1: Position and Orientation

Position in 3D Space

Position is defined using coordinates:

(x, y, z)

Example:

- Cube at (2, 1, 5)
- Chair at (10, 0, 3)

Position tells:

👉 Where object is located in world space.

Orientation

Orientation defines:

👉 Which direction object is facing.

It can be represented using:

- Euler angles (rotation around X, Y, Z)
- Quaternions (advanced, avoids gimbal lock)

Example:

- A door rotated 90° around Y-axis
- VR camera rotates as user turns head

Position = Location

Orientation = Direction

Transformations

Translation

Moves object from one location to another.

Example:

$(2, 3, 5) \rightarrow (5, 3, 5)$

Used when:

- Avatar walks
- Object moves

Rotation

Rotates object around axes:

- X-axis $\rightarrow$ pitch
- Y-axis $\rightarrow$ yaw
- Z-axis $\rightarrow$ roll

Used when:

- Head turns
 - Object spins
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Scaling

Changes object size:

- Uniform scaling (same in all directions)
- Non-uniform scaling (different per axis)

Example:

Scaling $(1,1,1) \rightarrow (2,2,2)$

PART 3: Bounding Volumes

Why Bounding Volumes?

Checking collision using full geometry is expensive.

Instead, we use simple shapes:

- Box
- Sphere

These are called bounding volumes.

They surround the object.

Types of Bounding Volumes

1. Bounding Box

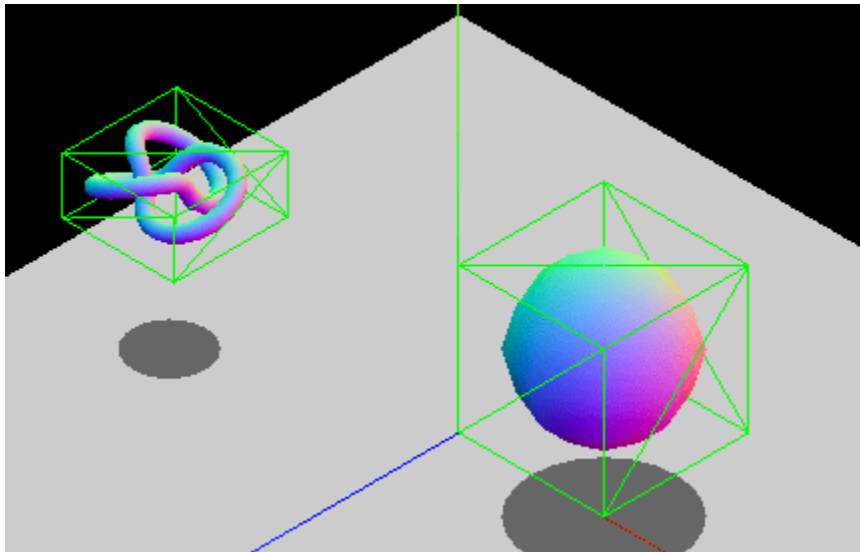
A **Bounding Box** is:

A rectangular 3D box that completely surrounds an object.

Instead of checking collision with a complex mesh (thousands of triangles), we check collision with a simple box.

This makes calculations much faster in VR.

Axis-Aligned Bounding Box (AABB)



Definition:

An Axis-Aligned Bounding Box (AABB) is a box that is aligned with the world coordinate axes (X, Y, Z).

It does NOT rotate with the object.

Even if the object rotates, the box remains aligned with the axes.

Key Characteristics:

- Edges parallel to X, Y, Z axes
- Easy to compute
- Very fast collision detection

- May include empty space when object rotates

Example:

If a car rotates:

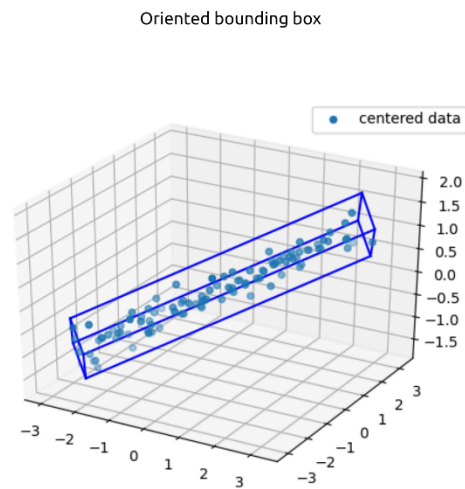
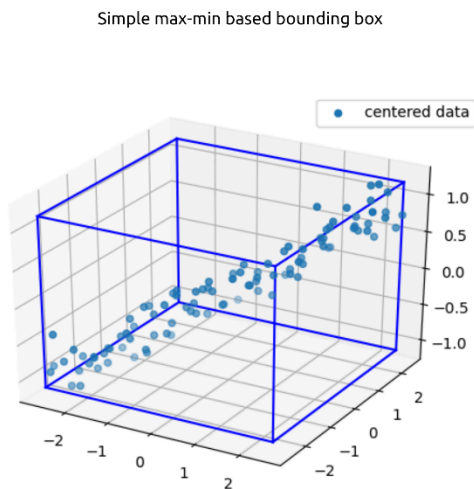
- The AABB stays axis-aligned
- The box becomes larger to contain rotated object

This can reduce accuracy but improves speed.

♦ When AABB is Used:

- Fast real-time VR applications
 - Broad-phase collision detection
 - Games requiring high FPS
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Oriented Bounding Box (OBB)



Definition:

An Oriented Bounding Box (OBB) is a bounding box that rotates along with the object.

It is NOT restricted to world axes.

Key Characteristics:

- Rotates with the object
 - More tightly fits object
 - More accurate collision detection
 - More computationally expensive
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Example:

If a sword rotates:

- OBB rotates with it
 - Maintains tight fit
 - Better for precision interaction
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AABB vs OBB Comparison

Feature	AABB	OBB
Alignment	World axes	Object orientation
Accuracy	Moderate	High
Speed	Very fast	Slower
Rotation handling	Expands when rotated	Rotates naturally
Use case	Games, VR apps	Simulation, precise collision

In VR Systems

Typically:

1. First check using AABB (fast)
2. If collision likely → use OBB or mesh-level check

This is called:

Broad-phase → Narrow-phase collision detection

Advantages:

- Simple
- Fast collision detection

Disadvantages:

- May not tightly fit complex shapes
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2. Bounding Sphere

A sphere enclosing the object.

Advantages:

- Very fast collision check
- Rotation independent

Disadvantages:

- Not accurate for irregular shapes
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Comparison Table

Feature	Bounding Box	Bounding Sphere
Shape	Box	Sphere
Accuracy	Moderate	Less for complex shapes
Speed	Fast	Very fast
Rotation handling	Complex (AABB)	Easy

Real VR Example

In VR shooting game:

- Bullet collision checked with bounding sphere

- Player body collision checked with bounding box
- Only if bounding volumes intersect → detailed check done

This improves performance.

Recap + Questions

Recap:

- Position = location in 3D
- Orientation = direction
- Transformations move objects
- Bounding volumes optimize collision detection

Quick Oral Questions:

- What is difference between position and orientation?
- Why are bounding volumes used?
- Which is faster: bounding sphere or full mesh collision?
- Compare AABB and OBB?